HACK YOUR PATH 7.0

Theme : Open Innovation (SDG-Alligned)

Problem Statement Title - CortexKey: Brainwave-Backed Authentication

Team Name : BlackHats

Team Details :

1. Devesh CSD 2nd yr
2. Aditya CSD 2nd yr
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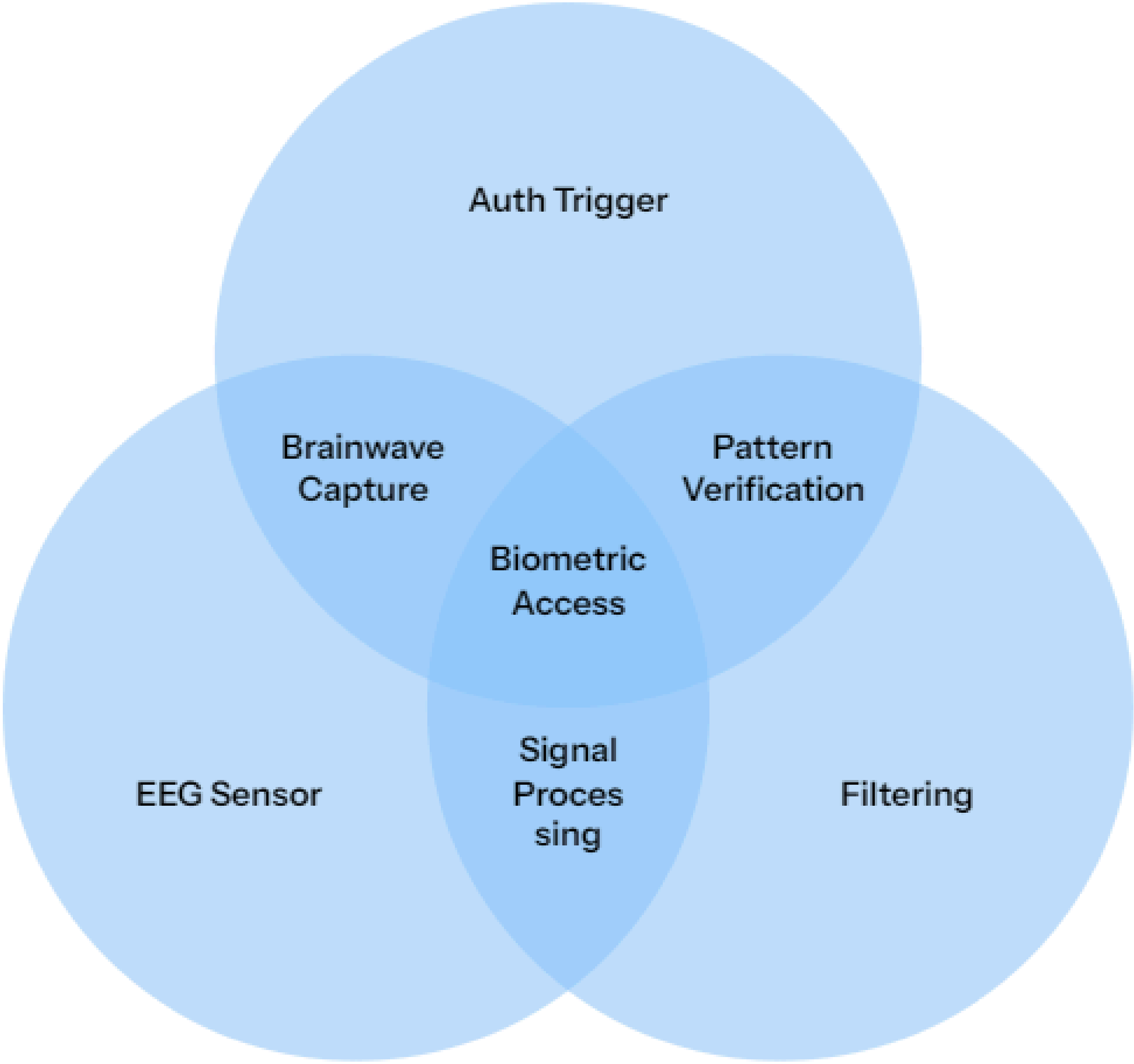
IDEA TITLE

Problem Statement

Current authentication methods fall short:

- Passwords are reusable, guessable, and vulnerable to phishing
- Biometrics like Fingerprint and FaceID rely on static physical traits
- Once compromised, biometric data cannot be changed
- Single-factor authentication increases the risk of unauthorized access



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Solution: Evolution of Authentication Systems

From static credentials to brainwave-based biometric login

CortexKey – Secure Login Using Brainwaves

- Traditional biometrics authenticate using physical traits
- These can be replicated or misused

CortexKey:

- Uses EEG brainwave features for login authentication
- Acts as an add-on or alternative to FaceID/fingerprint
- Ensures strong liveness-based identity verification

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

SUSTAINABLE DEVELOPMENT GOALS

16 PEACE, JUSTICE AND STRONG INSTITUTIONS

Future of Authentication

CortexKey is poised at the forefront of next-gen security, evolving with wearable tech.

Today

Proof of concept, rapid prototyping, and foundational architecture.

Near Term

Pilot programs in high-security environments and educational platforms.

Long Term

Integration into mainstream consumer BCI wearables and ubiquitous secure access.

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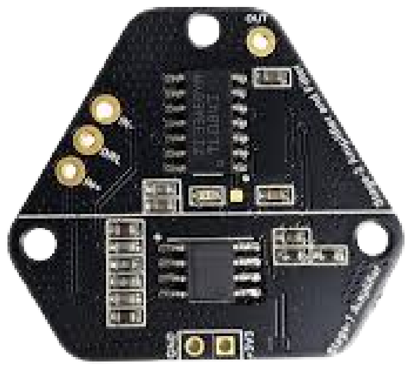
TECHNICAL APPROACH



Hardware Layer

Leveraging cutting-edge bio-sensing technology for accurate signal capture.

- **Sensor:** Neuphony EXG Synapse or BioAmp EXG Pill (Analog Front-End)
- **MCU:** ESP32 or Arduino Uno R3 (Data transmission)
- **Power:** Isolated DC source to minimize 50Hz mains noise

Software Layer

Efficient and reliable code for seamless data handling and system control.

- **Languages:** Python 3.8+ (Backend), C++ (Firmware)
- **Drivers:** Standard Serial/UART communication drivers

Processing Pipeline

A four-stage process transforming raw brainwaves into an authentication trigger.

1. **Acquisition:** Raw capture via Frontal Lobe electrodes.
2. **Preprocessing:** Digital 50Hz Notch Filter & 5Hz–30Hz Bandpass Filter.
3. **Extraction:** Power Spectral Density (PSD) analysis.
4. **Authentication:** Lightweight SVM Classifier matching.

ML & Libraries

Utilizing industry-standard tools for signal processing and machine learning.

- **Signal Processing:** MNE-Python, SciPy
- **Machine Learning:** Scikit-learn (SVM/Decision Tree)

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FEASIBILITY AND VIABILITY

Feasibility:

- Uses existing EEG signal processing methods
- Low-cost, non-medical EEG hardware
- Software-driven authentication logic
- Prototype buildable within hackathon time
- No dependency on large ML models

Viability

- Addresses real security gap in passkeys & biometrics
- Works as a second factor, not a replacement
- Scales with improving wearable EEG tech
- High relevance for secure systems & exams
- Strong future adoption in high-security domains

Item	Product	Approx. Price (INR)
BCI Sensor	**Neuphony EXG Synapse** OR **BioAmp EXG Pill**	₹1,200 – ₹1,500
Microcontroller	**Arduino Uno R3** (Clone) or **ESP32**	₹450 – ₹600
Electrodes	Gel-based disposable ECG/EEG studs	₹200 – ₹300
Cables	Bio-potential cables (Snap-on type)	₹300 – ₹400
Power	9V Battery	₹100
Total		**~₹2,500 – ₹2,900**

NOTE:
The price range for a standard hardware passkey falls between ₹3,000 and ₹20,000.

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RESEARCH AND REFERENCES

Academic Publications

- Chuang, J., Nguyen, H., & Sayed, B. (2013). "Cerebrem: A Brain-Computer Interface for User Authentication." Demonstrates the use of mental tasks (e.g., imagining movement) as a viable, secure cryptographic key.
- Gui, Q., Zhan, Z., & Zhu, J. (2014). "A Survey of EEG-based Biometrics for User Authentication." Validates brainwaves as high-entropy, non-static biometric identifiers superior to fingerprints.
- Ashby, C., et al. (2011). "Low-Cost EEG Based BCI User Identification." Proves that non-medical grade EEG sensors (like the EXG Pill) are sufficient for robust identity verification.
- Palaniappan, R. (2004). "Method of Identifying Individuals Using Brain Electrical Activity Features." The foundational study for using Power Spectral Density (PSD) as a "Neural Signature."

Technical Frameworks & Tools

- BrainFlow SDK: The industry-standard API for real-time biosensor data acquisition across open-source hardware.
- MNE-Python: Primary library for scientific-grade EEG signal processing, noise filtering, and artifact removal.

GTM Canvas

Project Name:	Cortex key
GTM Lead:	devesh(blackhats)
Date: 20-02-2026	Iteration # 1.0

1. Primary Feature

- **Dynamic Cognitive Verification**

Real-time extraction of Alpha/Beta EEG neural signatures using a lightweight edge-processing pipeline (SVM Classifier). Transforms mental tasks into cryptographic keys.

2. Primary Sellable Feature

- **Hardware-Backed Liveness 2FA** Continuous, deepfake proof authentication. It proves unequivocally that a conscious, living human is present at the terminal, neutralizing AI spoofing and static credential theft.

3. Differentiator

The World's First Resettable Biometric

Unlike fingerprints or FaceID , if a CortexKey signature is intercepted, the user simply changes their "mental task" to instantly generate a new, secure biometric key.

4* Alpha (unfair advantage)

- Extreme Cost-to-Precision Ratio (<₹3,000 BOM) Delivering enterprise-grade,
- statergiic innovation

5. Handpick Users (Atomic Network)

Name	Email	Contact	Degree
content creators youtube twiter			1st
HNI			2nd
fintech bank			1st
Crypto Exchange			2nd

8. Medium/Channel (how)

Live "Spoof-Proof" Demos (offering bounties to hack the headband), high-level technical whitepapers on "The Death of Static Biometrics", and direct B2B pilot programs offering our ₹3,000 dev-kit for free testing.

7. Source (where)

Where they congregate: High-level Cybersecurity Conferences (DEF CON, Black Hat, Nullcon), Identity Management Summits, Private LinkedIn CISO Guilds, and Deep-Tech Policy Forums.

6. Profiling (who)

social media and securities

these are the specific region with speciific problem which comes with first customer

Partnerships

DISCOVER

Inbound Organic

Inbound Organic: * SEO (Keywords: Resettable biometrics, Zero-Trust EEG, BCI Auth).
Open-Source GitHub Repos (publishing our BrainFlow integration to build developer trust).
Technical Blogging (Medium/Substack on "AI Deepfakes vs. Brainwaves").
Inbound Paid: * Niche LinkedIn Ads specifically targeting job titles: "CISO", "IAM Director", "Zero Trust Architect".

Inbound Paid

DISTRIBUTION

Outbound Paid

Outbound Paid: * Sponsored live demos at Tech/Security Expos.
B2B placements in Threat-Intel Newsletters.
Outbound Organic: * Direct Enterprise Piloting (sending hardware kits to security teams).
Guest Podcasting on DeepTech and Cybersecurity shows.
Product Tech Reviews (getting security influencers to try to "hack" the system on camera).

Outbound Organic